

Quality Assurance Department
Blueberry Quality Grading Standards and
Inspection Procedures for DEMEA Blueberries

Section I:	Specification Narrative
Section II:	Process of Score Calculation and Fruit Grading
Section III:	Weighted Factors for Scoring Blueberries
Section IV:	Defect Tolerances
Section V:	Minimum Package Weights
Section VI:	Lot code
Appendix I:	Photo Fact Sheet Defects/Color

Section I: Specification Narrative

Quality/Appearance product description	Driscoll's blueberries should be picked fully ripe and packed with care—they should exhibit mature flavor and color at the time of harvest. Berries should have an attractive, uniform appearance: medium to light blue skin color, medium to large fruit size and nearly spherical shape. Flavor should be sweet with palatable acid balance and no bitterness or off-flavor. Berries should have firm but yielding flesh texture—never mushy or mealy— without gritty seediness or tough skin. Berries must be fresh in condition: plump, intact with no physical damage or decay and completely separate from one another; not clustered. Surface contamination including dust, minerals, must be absent. Finished packs should not contain extraneous material, including blossoms, stems, leaves, water or insects. The most delightful berries in the pack will be juicy, crisp and have characteristic “blueberry flavor”.
Transport/Storage temp	2°C
Shelf life	At least 15 days
Stacking pallets	Composition of the pallets is preferably done on variety

- The quality of the blueberries will be determined on batch level. The size of the batch depends on the use of batch attributes. Examples of batch attributes are: Grower/Supplier, Ranch, Field, Block, Harvest date, Fruit type, Same delivery, QC score, L-code, Pallet type. Per pallet 0,18% of the punnets is checked and samples will be taken randomly. All the found defects during intake will be placed in 2 two separate groups: Condition and Appearance. In the table below the calculation method is explained step by step.

For example 500 blueberries are counted in the sample. Of those 500, 5 are leaky and 5 are picked to green

Test Condition		Test Appearance
Number of times defect appears X Weight Factor = Answer 1 (Calculate per defect)		Number of times defect appears X Weight Factor = Answer 1 (Calculate per defect)
Result of Answer 1 divided by The total fruit count = Answer 2		Result of Answer 1 divided by The total fruit count = Answer 2
100 - Answer 2 = Answer 3 PFQ Condition		100 - Answer 2 = Answer 3 PFQ Appearance X
Answer 3 PFQ Condition * 3 + Answer 3 PFQ Appearance *1 / 4 = Final PfQ Score		

$$\text{PfQ blueberries} = ((3*\text{Cond}) + (1*\text{App})) \text{ or } \rightarrow \text{PfQ} = \frac{(3*\text{cond}) + (1*\text{app})}{4}$$

Because $\frac{3}{4} = 0.75$ and $\frac{1}{4} = 0.25$

5 Leaky x 2000 = 10.000 (Answer 1)		5 Green x 800 = 4000 (Answer 1)
10.000 (Answer 1) divided by 500 blueberries = 20 (Answer 2)		4000 (Answer 1) divided by 500 blueberries = 8 (Answer 2)
100 – 20 (Answer 2) = 80,00 Condition score (Answer 3)		100 – 8 (Answer 2) = 92,00 Appearance score (Answer 3)
<i>The outcome of answer 3 (condition or/and appearance) cannot score lower than zero in this calculation.</i> <i>Calculate the average of those two scores:</i> 80,00 *3Condition score (Answer 3) + 92.00 *1 Appearance score (Answer 3) / 4 = 83,00 PfQ score		

Section III: Weighted Factors for Scoring Blueberries

- The following table illustrates the weighted percentages per defect.

Outcome	Description	Factor (%)	Test
Wet / Leaky	Wet / Leaky / Overripe	2000	T2 Blue Condition
Rot / Mould	Mould / Rot / Collapsed	5000	T2 Blue Condition
Soft / Shriveled	Shriveled / Sensitive/Soft	800	T2 Blue Condition
Larvae <1%	Presence of Larvae	10000	T2 Blue Condition
Pest damage	Pest or Insect damage	1000	T2 Blue Appearance
Skin defect	Abnormalities in skin, Cracked skin	600	T2 Blue Appearance
Foreign bodies	Dust / Dirt / Sand / Foreign bodies-/ Feces	1000	T2 Blue appearance
With Stem	With Stem	400	T2 Blue Appearance
Size	<12 mm	600	T2 Blue Appearance
Green / Unripe	Green / Unripe	1600	T2 Blue Appearance
Bloom	Absence of Bloom	1000	T2 Blue Appearance

Section IV: Defect Tolerances

Defects/Requirements	Driscolls Packed Fruit Percentage Allowed pfq100-70	Driscolls Packed Fruit Percentage Allowed Pfq 69-50	Blanco Packed Fruit Percentage Allowed Pfq 49-25	Blanco Packed Fruit Percentage Allowed Pfq 25-0
Mould	Up to 1%	Up to 1%	Up to 2%	More than 2%
Larvae	0%	0%	Up to 1%	Up to 1%
Size	Min. 12 mm.	Min. 12 mm.	Min. 12 mm.	Min. 12 mm.
Size Food Service	Min. 16 mm.	N.A.		

Section V: Minimum Package Weights

- For all DEMA packing a minimal percentage as mentioned below on top of the net fruit weight is required. It is illegal to ship underweight product, therefore if any clamshell in a pallet is found to be underweight, pallet(s) will be sent to repacking.

	Blueberries
Spain	3%
Portugal	4%
Morocco	5%
NBGF	2.5%
South America	5%
South Africa	5%
Serbia	5%

Section VI: Lot code

- Lot code (EU 2011/91)

Within EU, it is obligated to add an L code for identification on a batch. Within Driscoll's each punnet is provided with a lot code. Punnets contain a lasered lot code on the bottom of the punnet or a sticker which is the same as the box lot code. See examples below:

Driscoll's Standard
Punnet: lasered
Box: wk + day + grower

Example 'Punnet: lasered'



Example "Wk+Day+Grower+Variety or
Block
For Punnet & Box



Week 16, Wednesday, grower 339, Sweet Jane

Monday=1 Tuesday=2 Wednesday=3 Thursday=4 Friday=5 Saturday=6 Sunday=7

Variety number according Driscoll's variety overview list

- Position of the L-code sticker (if applicable)

L-code should be placed on the bottom of the punnet, next to the bottom label.

L-code on the box should be placed in the center of the shortest side of the box



Appendix: Photo Fact Sheet Defects and Color



Appendix: Photo Fact Sheet Defects and Color



Appendix: Photo Fact Sheet Defects and Color



Appendix: Photo Fact Sheet Defects Color





More than 5 mm of color deviation
(green/ red) will not be tolerated;
less than 5 mm will be accepted



